

Separating Mixtures

Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Recovering materials

Name: _____ Date: _____

LEARNING INTENTION & SUCCESS CRITERIA

I can choose a separation method using properties of materials.

I can match a method to a mixture, explain why it works and plan how to recover a dissolved solid.

METHOD HELPER

Hand-picking: remove visibly different pieces.

Sieving: separate solid pieces of different sizes.

Magnetic separation: remove a magnetic material.

Filtration: separate an insoluble solid from a liquid.

Evaporation: recover a dissolved solid as the solvent evaporates.

WORKED EXAMPLE

Sand + water: filter the mixture. The sand is trapped by the filter; water passes through. Dissolved salt passes through an ordinary classroom filter with the water.

PART A - CHOOSE A METHOD

Write a suitable method and the property that makes it work.

1. Large pebbles mixed with fine sand.

2. Steel paper clips mixed with dry rice.

3. Sand mixed with water.

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Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Apply & reason

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PART B - CORE SKILLS

4. Salt is dissolved in water. Choose a method to recover the salt. Why will an ordinary paper filter not work?

5. A student filters muddy water and says, "It is safe to drink now." Why is this conclusion unsafe?

PART C - SEPARATION CHALLENGE

A mixture contains sand, salt and water. The salt has dissolved.
Plan two steps to recover the sand and then the salt. Label what is collected at each step.

6. Step 1: method, material collected and reason.

7. Step 2: method, material collected and reason.

REFLECTION & I CAN CHECK

The property I used to choose a method was: _____

I can explain filtration: Yes / Not yet

I can explain salt recovery: Yes / Not yet

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Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Parent / teacher answers

Name: _____ Date: _____

ANSWER KEY

1. Sieving: fine sand passes through suitably sized holes; large pebbles remain. Hand-picking is acceptable for a small sample.
2. A magnet attracts the steel clips but not rice. Hand-picking is also possible; credit a justified method.
3. Filtration: insoluble sand grains are retained and water passes through.
4. Evaporation: water leaves as vapour and salt remains. Dissolved salt is not trapped by an ordinary classroom paper filter.
5. Filtration does not necessarily remove microorganisms or dissolved harmful substances. The water must not be drunk.
6. Filter first. Collect sand on the filter; collect saltwater below it.
7. Evaporate water from the collected saltwater to recover salt. Sand and salt are recovered separately; this method does not collect the water.

TEACHING & PARENT TIP

Use clean sand, water and a coffee filter for a demonstration. Keep all experiment mixtures out of mouths. Recover salt by natural evaporation in a labelled shallow dish; allow several days if needed.

The key question is “Which property makes your method work?” These introductory techniques support Year 6 reversible-change learning; the broader separation topic is developed directly in Year 7.

TEACHER FEEDBACK

Strength: _____

Next step: _____